

OPINION

To differentiate or not — routes towards metastasis

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Abstract | Why are many metastases differentiated? Invading and disseminating carcinoma cells can undergo an epithelial–mesenchymal transition (EMT), which is associated with a gain of stem cell-like behaviour. Therefore, EMT has been linked to the cancer stem cell concept. However, it is a matter of debate how subsequent mesenchymal–epithelial transition (MET) fits into the metastatic process and whether a MET is essential. In this Opinion article, I propose two principle types of metastatic progression: phenotypic plasticity involving transient EMT–MET processes and intrinsic genetic alterations keeping cells in an EMT and stemness state. This simplified classification integrates clinically relevant aspects of dormancy, metastatic tropism and therapy resistance, and implies perspectives on treatment strategies against metastasis.

Metastasis is of the utmost clinical relevance, as it is responsible for more than 90% of cancer-associated mortality¹. Therefore, the clinical need to prevent metastasis formation or to target existing metastases is substantial. The best way of developing novel therapeutic strategies is to understand the biology that underlies metastasis.

A basic observation in distant metastases in all types of epithelial cancers (carcinomas) is that a high proportion of them are differentiated, and in some cases metastases can show a greater degree of cellular differentiation than the corresponding primary tumour. Indeed, this is typical for metastases of well differentiated to moderately differentiated adenocarcinomas, which express the same glandular morphology as their primary tumours. At first glance this seems trivial; however, it is astonishing when one considers that cancer cells must disseminate through a fine ‘net’ of blood vessels, which would obviously be a difficult task for epithelial-differentiated tumour cell clusters. In addition, both differentiated primary tumours and corresponding metastases often have a similar heterogeneous organization, which is characterized by regions of dedifferentiation, particularly at the invasive front (FIG. 1).

This dedifferentiation shows hallmarks of an epithelial–mesenchymal transition (EMT), and it is now accepted that the acquisition of an EMT in differentiated cancers can strongly enhance tumour cell dissemination². On the basis of findings in colorectal cancer metastases, we initially proposed transient EMT–MET processes as the underlying driving forces of metastasis³. In this model, a dedifferentiation resembling an EMT with a loss of E-cadherin was detected in invasive cancer cells of primary tumours, and a reversal of this undifferentiated phenotype (a MET) that was characterized by the re-expression of E-cadherin was seen in corresponding liver metastases. Further analyses of gene expression patterns in colorectal adenocarcinomas and their corresponding liver metastases indicated that invasive, dedifferentiated cancer cells combine EMT properties with a stem cell-like phenotype. This led to the concept that these cells combined traits that are necessary for acquiring a motility and a stemness phenotype and, therefore, we termed these cells ‘migrating cancer stem cells’ as potential sources of metastases⁴. This concept was supported by the important experimental finding that the induction of an EMT can co-induce stem

cell properties, thereby coupling cell motility and stem cell-like programmes^{5,6}. The biological and clinical consequences of these findings are far-reaching, as cancer cells, with an aberrantly activated EMT programme, receive all the necessary traits for dissemination and metastatic seeding in one go. The ‘classical’ EMT properties induce aberrant cellular motility². The classical stemness properties, including apoptosis resistance, transient quiescence and self-renewal capacities, allow survival during dissemination, colonization at the metastatic site, eventual drug resistance and long-term maintenance of cancer stem cells. Further characterization of cancer stem cells in different tumour types is required to show whether EMT and stemness properties are always linked. This cascade of discoveries allows the merging of the cancer stem cell theory⁷ with the EMT–MET concept and results in a comprehensive hypothesis of abnormal phenotypic plasticity that enables the permanent adaptation of cancer cells to the challenging changes in the tumour environment. Although many clinical reports foster this concept of transient EMT–MET processes in metastasis, there remains little experimental proof that it is correct. However, many experimental results and conceptual advances support the role of aberrant phenotypic plasticity as one of the driving forces for metastasis.

Conversely, it has long been known that undifferentiated metastases also occur in cancer patients. Even in an individual patient, heterogeneity in the differentiation status of the metastases is possible, as multiple metastases in one organ may be differentiated and undifferentiated, as can be seen in colorectal, breast and lung cancer metastases⁸. The differentiation state of metastases is also associated with clinical outcome, as demonstrated for unresectable liver metastases of colorectal cancer, in which a low level of differentiation correlates with a poor 2-year survival rate^{9,10}. It seems that undifferentiated metastases have not undergone, and do not need, a redifferentiation or MET on colonizing their secondary site. There are several possible explanations for this that are based on genetic changes rather than on phenotypic plasticity.

In this Opinion article, I discuss the role of cellular plasticity as the crucial motor for

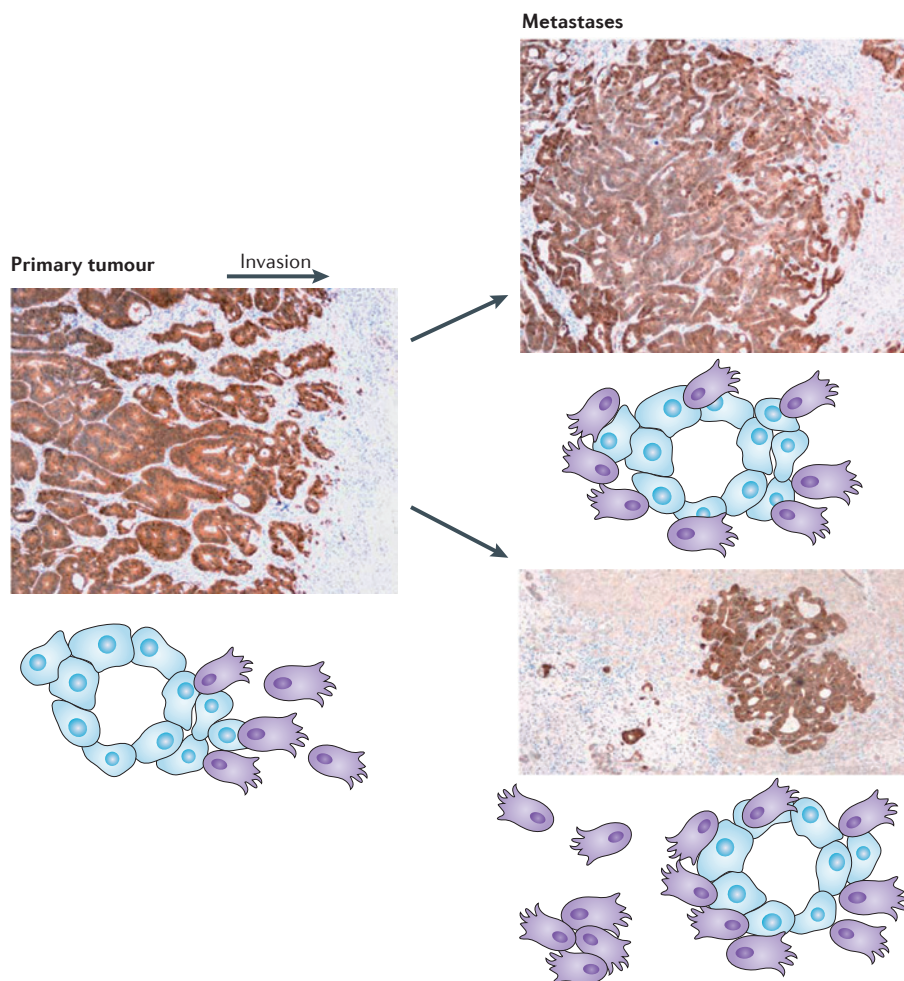


Figure 1 | Examples of typical differentiated metastases. Histology of a typical well-to-moderately differentiated colon adenocarcinoma and two corresponding liver metastases that are stained for the expression of β -catenin (brown) is shown. Note that the primary tumour and metastases show the same heterogeneity, with a tubular differentiated phenotype in the centre and an undifferentiated phenotype at the periphery. This is also indicated schematically (a differentiated phenotype is indicated with blue and an undifferentiated phenotype is indicated with purple). A small differentiated metastasis with adjacent single, undifferentiated tumour cells is shown (bottom panel).

the metastasis of differentiated carcinomas, and link it to central aspects of metastasis, including dissemination, dormancy, colonization, organ tropism, metastatic niches, therapy resistance and subsequent clinical perspectives (BOX 1). I do not neglect the existence of undifferentiated metastases, but instead integrate genetic progression as an additional driving force, resulting in a simplified classification of two principle types of metastatic progression that are based on phenotypic plasticity and/or on intrinsic genetic alterations.

Routes to metastasis

The most important steps for distant metastasis are dissemination through a fine net of blood vessels and colonization at the metastatic site. In contrast to undifferentiated,

anaplastic primary tumours, cells from differentiated tumours are not expected to possess the necessary traits with which to disseminate, nevertheless they also metastasize. Therefore, on the basis of observations from clinical samples, the proposal of transient rounds of EMT–MET resulting in abnormal phenotypic plasticity is a straightforward concept that explains the metastasis of differentiated cancers. But, in order to be comprehensive, a discussion about routes to metastasis must include the fact that metastases can also be undifferentiated. Therefore, I propose a simplified classification of two principle types of metastasis formation (FIG. 2).

In plasticity type I, metastases have a differentiated phenotype and retain the hierarchical organization of the primary tumour. They have undergone transient transition

processes, starting with an EMT that enables invasion, dissemination and seeding. For colonization and formation of macrometastases, a MET and redifferentiation are necessary. Thus, the process is characterized by high phenotypic plasticity, which is mostly triggered and regulated by environmental conditions and contextual signals. Reversible epigenetic modifications rather than fixed genetic alterations in relevant genes are suggested as an underlying molecular basis.

Genetic type II metastases have an undifferentiated phenotype, irrespective of their primary tumour, which can be differentiated or undifferentiated. Invading and disseminating tumour cells are in a permanent EMT and stemness-like state, and only weak redifferentiation is possible and/or necessary to form macrometastases. This state is generally irreversible owing to tumour cell intrinsic properties and the accumulation of genetic alterations, and thus the differentiation capacity is low and the overall phenotypic hierarchy is flat.

There are important common and different principles for both types of metastases. Without doubt, founding genetic alterations, such as *APC* gene inactivation in colorectal cancer, are the basis for the initiation of cancer, irrespective of the dominant mechanisms that result in metastatic progression. However, for plasticity type I metastasis, these initiating mutations could result in aberrant reactions to external stimuli, resulting in enhanced cellular plasticity. The most important common molecular trait is that both types of metastasis rely on invading and disseminating tumour cells, probably in a combined state of EMT and stem cell-like capacities. The combination of both traits is considered to strongly enhance successful metastasis formation¹¹. The major differences between plasticity type I and genetic type II are the cause or trigger of the EMT–stem cell-like state and its potential reversibility as a pre-condition to form macrometastases.

Plasticity type I

Molecular basis of plasticity. The most important principle underlying type I metastasis is that the EMT (and therefore a stem cell-like state) is transient and reversible. But what are the underlying molecular changes that enable cellular plasticity and adaptation to different environmental challenges, and how are these processes controlled? Recent results have shown that EMT-inducing transcriptional repressors have an important role and often involve reciprocal interactions with microRNAs. For example, a series of

publications has indicated that cellular plasticity is exerted by a reciprocal feedback loop between the ZEB family of EMT inducers (ZEB1 and ZEB2) and the miR-200 family as an inducer of epithelial differentiation^{12–17}. Within this ZEB–miR-200 feedback loop, ZEB inhibits the transcription of miR-200 family members, and miR-200 family members inhibit the translation of ZEB, thus both factors control the expression of one another (FIG. 3). ZEB1 induces EMT and a stem cell-like state not only by directly inhibiting the expression of epithelial proteins, but also by repressing its own repressor miR-200. Most importantly, miR-200 induces differentiation not only by targeting its own repressor ZEB1, but also by directly inhibiting the translation of stem cell factors and stem cell-associated epigenetic regulators, such as BMI1 (REFS 18, 19) and SUZ12 (REF. 20). The potential clinical consequences are far-reaching. ZEB1 is a strong inducer of tumour cell invasion and is necessary for metastasis in animal models^{21,22}. It is overexpressed in a large number of human cancer types and is associated with poor prognosis²³. Intriguingly, miR-200 can be overexpressed in certain cancer types, such as in ovarian, endometrial and pancreatic cancer, and its expression level is also associated with poor prognosis^{24–27}. One molecular explanation for these contradicting findings is that, although miR-200 downregulation

enhances dissemination, its re-expression, by inducing a MET, is crucial for metastatic colonization and macrometastasis formation²⁸. The ZEB–miR-200 feedback loop also has a central role in controlling drug resistance. ZEB1 expression has been shown to confer resistance to epidermal growth factor receptor (EGFR) inhibitors and standard chemotherapeutics, such as gemcitabine^{19,29–32}. By contrast, drug sensitivity can be restored by overexpression of miR-200 family members^{19,33–36}.

A weak point in the proposed model of the ZEB–miR-200 feedback loop as a driver of phenotypic plasticity is that other potent EMT inducers, such as SNAIL1 (also known as SNAIL), are not targeted by miR-200 and, therefore, are not directly controlled by this feedback loop. This problem was recently resolved by the demonstration that SNAIL1 is embedded in a second reciprocal feedback loop with miR-34, which, remarkably, follows the same principles (FIG. 3): SNAIL1 inhibits the transcription of miR-34 family members and miR-34 inhibits the translation of SNAIL1 (REFS 37, 38). Moreover, the SNAIL side of the loop induces EMT, stem cell-like characteristics and drug resistance, and the miR-34 family induces MET, differentiation and drug sensitivity. Strikingly, gene loci of both miR-200 and miR-34 family members are frequently inactivated by epigenetic mechanisms in different model systems

and undifferentiated regions of human cancers^{39,40}. As epigenetic inactivation is potentially reversible, such data support the suggestion that epigenetic regulation rather than mutation could be the basis for transient phenotypic changes and cellular plasticity underlying type I metastasis, and these data also point to novel therapeutic options (discussed below). However, one of the most remarkable and important discoveries concerns the regulation of both feedback loops. Whereas much is known about the extracellular and intracellular control of EMT inducers, little was known about what activates the expression of both miR-200 and miR-34. Strikingly, transcription of both microRNA families is induced by p53 (REFS 41–44), placing one of the most important tumour suppressors centre stage for the regulation of phenotypic plasticity⁴⁵. The relevance of this finding for cancer biology is high, as it implies that normal p53 function is required for phenotypic plasticity. Consequently, are p53 mutations a way to genetically fix cancers in an EMT and stemness state? This idea is supported by findings that epigenetic inactivations of *mir-34* genes and p53 mutations are mutually exclusive in colorectal cancers⁴⁰. However, the generally high abundance of p53 mutations in many cancers, as well as in differentiated types, suggests that p53 mutations alone cannot be sufficient for a permanent EMT and stemness state. It will be necessary to investigate whether p53 mutations are cooperating with other known or unknown genetic alterations to fix this state in undifferentiated tumours and their metastases.

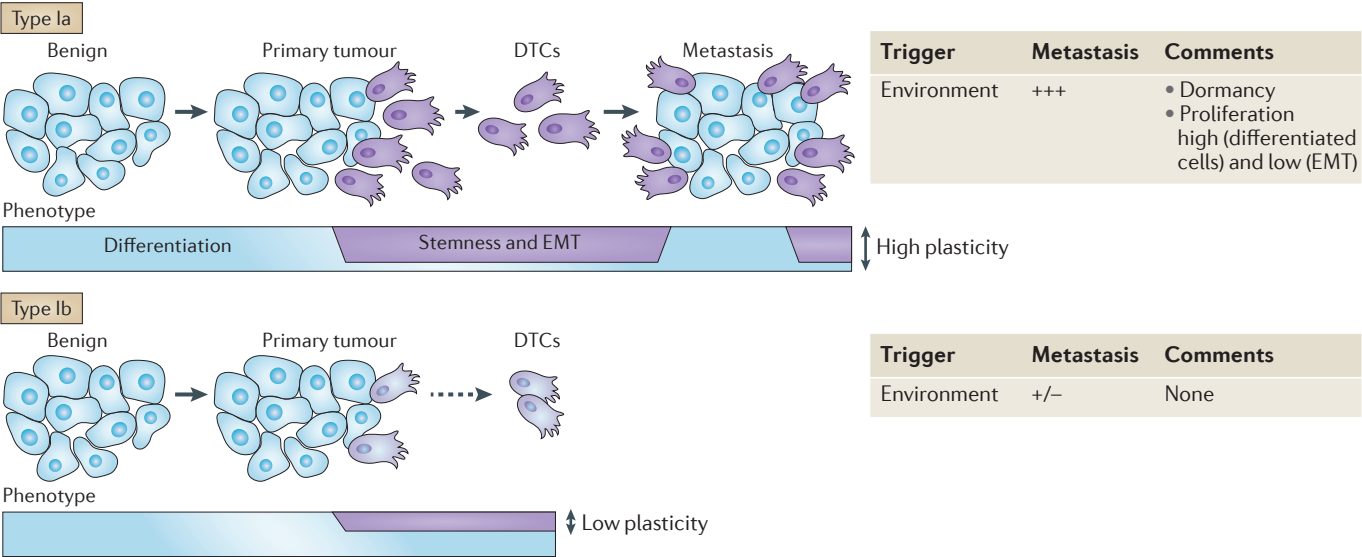
Why do metastases re-differentiate?

Although there are likely to be several answers to this question, one likely answer is that in differentiated tumours the capacities to grow and to disseminate are mutually exclusive. In our initial report that suggested that transient EMT–MET processes are the driving force of metastasis in differentiated cancers, we realized that invasive colorectal cancer cells that had undergone an EMT expressed low levels of the proliferation marker Ki67 (REF. 3). Strongest proliferation was seen in the differentiated regions of both the primary tumour and the metastases, and, therefore, we suggested that a MET or re-differentiation is necessary to overcome EMT-associated growth arrest. This was further supported by showing that invasive cells express the cell cycle inhibitor and senescence marker INK4A (also known as p16)^{46,47}. Notably, the abundance of INK4A in invasive cancer cells correlated with poor

Box 1 | The biology of distant metastasis at a glance

The ‘metastatic cascade’ starts with the invasion and dissemination of cancer cells from the primary tumour (or possibly from its precursor lesion¹⁴²). It is now accepted that aberrant activation of an epithelial–mesenchymal transition (EMT)–stemness programme, which is triggered by environmental factors such as inflammation and hypoxia, is a major driver of the abnormal motility of cancer cells¹¹. This is associated with the activation of genes, the products of which enhance the competence for metastasis at all levels of the metastasis cascade, the so-called metastasis virulence factors¹⁴³. In total, this enables migrating cancer stem cells (CSCs) to enter blood vessels, disseminate in the body as circulating tumour cells (CTCs), seed into distant organs, form micrometastases and finally colonize to macrometastases^{1,2,4}. A related pool of CTCs known as disseminating tumour cells (DTCs) settles in the bone marrow. They can persist for various time periods in a stem cell-like state as dormant cancer stem cells and may be responsible for extreme differences in the latency until disease recurrence or the growth of metastasis, which can take up to 20 years in some patients with breast cancer^{77,144}. At the molecular level, dormant cancer cells can be in a state of quiescence or senescence⁵⁵. The rate-limiting step in metastasis is colonization — many CTCs can seed in experimental models of metastasis, but only a few are able to colonize, initiate growth and form macrometastases^{56,57}. Thus, understanding the colonization process is of the utmost relevance. Accumulating data indicate that the environmental conditions, generated in a crosstalk between the target organ, infiltrating cells and the primary tumour are crucial. One current theory is that systemic factors, secreted by the primary tumour, attract mesenchymal stem cells (MSCs) from the bone marrow to distant organs to form a pre-metastatic niche, which then allows seeding of CTCs. Additional crosstalk shapes the metastatic niche, allowing growth, vascularization and colonization of CTCs⁹¹. Thereby, a mesenchymal–epithelial transition (MET), allowing a re-differentiation of disseminated tumour cells may be a crucial process for macrometastasis of many differentiated carcinoma types. These only partially understood processes could also explain why certain tumours preferentially metastasize to certain target organs, a fact termed metastatic organ tropism. CTCs or DTCs will only colonize a metastatic niche if the right signals are present in the tissue¹⁴³.

a Type I plasticity



b Type II genetic

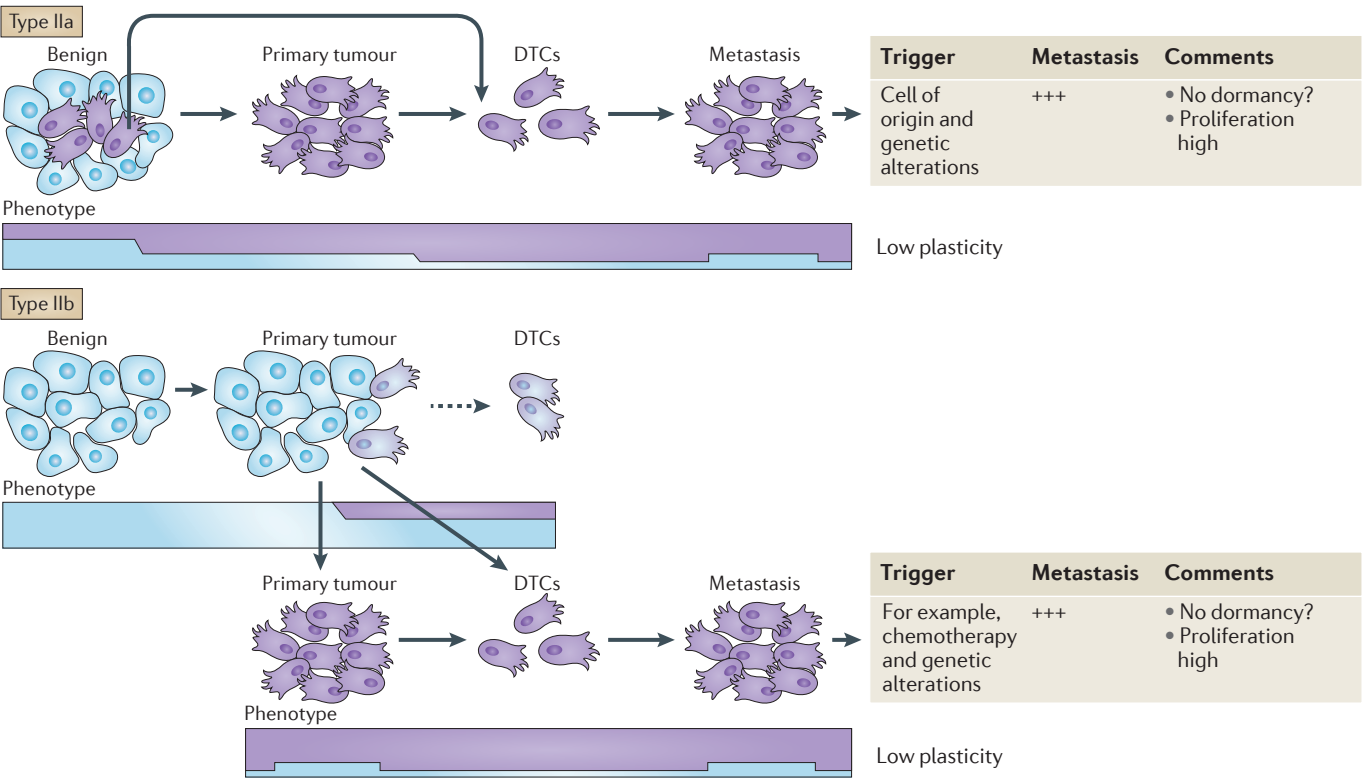


Figure 2 | The classification of metastasis in plasticity type I and genetic type II. a | Plasticity type I metastasis is characterized by re-differentiated metastases (shown in blue) and a transient loss of epithelial differentiation resulting in an epithelial–mesenchymal transition (EMT)–stem cell-like (stemness) phenotype (shown in purple). The EMT–stemness phenotype is probably associated with quiescence, or even dormancy, whereas differentiation allows growth. Depending on the possible range of plasticity, the metastatic capacity of differentiated primary tumours with the same grading may be high (type Ia, high plasticity) or low (type Ib, low plasticity). It is not understood why some, but not all (shown by the dashed arrows), tumours with the same differentiation status (or histological grading) are highly plastic. **b |** Genetic type II metastases are characterized by undifferentiated cells. They can be derived from intrinsically undifferentiated

tumours in an almost fixed EMT–stemness state, which never has the capacity for high differentiation, thus allowing only a flat hierarchy. Benign precursor lesions might not exist or might already be a source of disseminating tumour cells (DTCs) and metastases, thus allowing an early parallel progression of the primary tumour and metastases (genetic type IIa). Another source of undifferentiated metastases is plasticity type I cancers (either type Ia or Ib), which acquire additional genetic alterations, precluding differentiation and, therefore, lose their phenotypic plasticity. A clinically relevant selection pressure for such genetic alterations is likely to be long-term chemotherapy resulting in a drug-resistant EMT–stemness phenotype (genetic type IIb). Of note, in the concept of genetic type II, EMT and stemness and quiescence are decoupled, allowing permanent proliferation and thus might also preclude a dormancy phenotype.

prognosis in patients with differentiated colorectal cancers. Growth arrest is also an attribute of many normal tissue stem cells, as well as of many circulating tumour cells (CTCs) and disseminating tumour cells (DTCs). For example, it was shown that a large portion of bone marrow DTCs have stem cell properties and are in a quiescent state. These cells are considered to be dormant and persisting cancer cells, and their residence in the bone marrow is thought to keep them in a G0/G1 arrested state^{48,49}.

The cell cycle arrest that is evident in invading and disseminating tumour cells can also be explained at the molecular level. It has long been known that the induction of an EMT by transforming growth factor- β (TGF β) is associated with reduced proliferation and growth arrest in epithelial cells⁵⁰, which can be executed, for example, by the EMT inducer ZEB1 (REF. 51). Vega *et al.*⁵² first demonstrated that SNAIL1 can directly induce a growth arrest by inhibiting the expression of cyclin D2 (REF. 52). This finding was extended by showing that SNAIL also directly suppresses proliferating cell nuclear antigen (PCNA) expression⁵³. Also, ZEB2 inhibits cell proliferation by targeting cyclin D1 expression and inducing phosphorylation of RB, which leads to a G1 arrest⁵⁴.

The requirement for a switch from a stem cell and dissemination-associated growth arrest to a proliferative state for the establishment of macrometastases is corroborated by several lines of evidence. First, DTCs can persist in patients with breast cancer as quiescent micrometastases for years after the removal of the primary tumours⁵⁵. The transition from micrometastasis to growing macrometastasis requires a switch from cell cycle arrest to proliferation. Most importantly, this transition, termed colonization, is thought to be the final rate-limiting step in metastasis. It was experimentally demonstrated that, although most CTCs survive the dissemination and seeding process, only about 0.01% of the tumour cells in systemic circulation are able to colonize and develop into macrometastases^{56,57}. It was recently shown that colonization is strongly enhanced by re-expression of miR-200 family members and subsequent epithelial differentiation²⁸. Of note, miR-200 expression also promotes the proliferation and growth of cancer cells⁵⁸. In addition, colonization requires an angiogenic switch for the blood supply of growing metastases^{59,60}.

In summary, these data indicate that the switch from migrating cancer stem cells or dormant and quiescent DTCs to growing cancer cells in macrometastases is coupled to an induction of differentiation. Thus, the

induction of a MET might be a rate-limiting and an important step in the metastasis of differentiated primary tumours. This also implies that cellular plasticity (the capacity to switch between an EMT–stem cell-like state and a differentiated state) is crucial for the formation of differentiated type I metastases. Therefore, at the molecular level, the feedback loops between ZEB and miR-200 and SNAIL and miR-34 may be important motors for exerting phenotypic plasticity and type I metastasis.

Evidence for plasticity. The existence of differentiated metastases is a clinical fact. However, it is still a matter of some debate⁶¹ whether EMT exists in carcinomas, with this question mostly triggered by the incorrect view that an EMT in a cancer cell is equal to the complete transition towards a pure mesenchymal phenotype, as is seen during development. One alternative hypothesis to exclude the need for an EMT in metastasis is that E-cadherin-expressing cells of the primary tumour are supported by dedifferentiated cells during dissemination, and that metastases are directly derived from these differentiated E-cadherin-expressing cells⁶². However, given the obstacles facing a disseminating tumour cell it is difficult to see how a differentiated epithelial cell could achieve this. Indeed, it was recently shown that tumour-initiating cells from xenografts of human colon cancers can only form liver metastasis if they have stem cell and self-renewal capacities⁶³. Moreover, phenotypic plasticity is not restricted to tumour cells, as it is particularly important during embryonic development. For example, an EMT induces gastrulation, and this is followed by a MET at implantation to form the trophoectoderm, the first embryonic epithelium⁶⁴. Phenotypic plasticity that involves rounds of EMT and MET is also central to tissue repair, and its aberrant activation is evident in pathological processes, such as organ fibrosis⁶⁵.

For almost 20 years it has been known that early stage colorectal cancers that invade with a budding phenotype have a worse prognosis than non-budding cancers⁶⁶. This was long before it was known that budding cancer cells have an EMT and a stem cell-like phenotype^{67,68}. High EMT in the invasive regions of differentiated primary tumours correlates with adverse clinical outcome and poor survival in early stage colorectal cancer^{66,69–71}, and additional studies have shown that the poor prognosis that is associated with budding-type colorectal cancers is due to metastases in lymph nodes⁷², as well as to distant metastasis to the liver^{68,73}

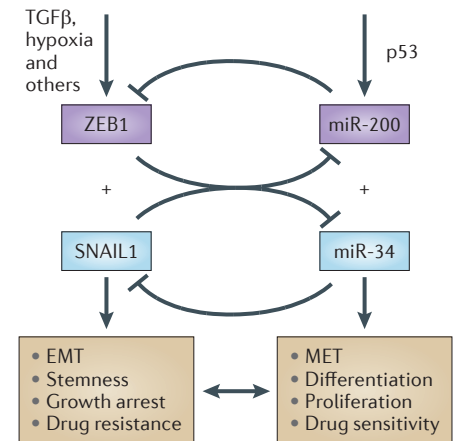


Figure 3 | Two reciprocal feedback loops exert phenotypic plasticity. Cytokines and certain extracellular conditions, such as hypoxia, stimulate the expression of epithelial–mesenchymal transition (EMT) activators of the ZEB family and SNAIL family, which induce EMT-associated cell motility, stemness, growth arrest and survival. These EMT activators are linked in double-negative feedback loops to mesenchymal–epithelial transition (MET)-inducing miR-200 and miR-34 family members. These EMT activators directly inhibit the transcription of the microRNAs (miRNAs) and, vice versa, the miRNAs block the translation of their inhibitory EMT inducers. Of note, miR-200 and miR-34 can shift tumour cells from a drug-resistant to a drug-sensitive phenotype. Thus, these feedback loops are the motor of phenotypic plasticity, enabling switches between the two states. Importantly, p53 activates the expression of both miRNA families, thereby shifting the feedback loops towards a MET, epithelial differentiation, proliferation and drug sensitivity in cancer cells. TGF β , transforming growth factor- β .

or the lung⁷⁴. Of note, the corresponding distant metastases were again differentiated. In breast cancer, the proportion of CD44⁺CD24^{low} cancer stem cells in the primary tumour correlates with increased risk of distant metastasis, and, strikingly, metastases seeded from these tumours sometimes show higher differentiation rates compared with the primary tumour, as indicated by increased expression of CD24 (REF. 75). In patients with metastatic breast cancer, the number of CTCs expressing EMT and cancer stem cell markers is strongly increased⁷⁶. Moreover, the number of DTCs correlates with poor prognosis⁷⁷. Breast cancer DTCs in the bone marrow show an increase in the proportion with a CD44⁺CD24^{low} stem cell phenotype^{48,78}, are non-proliferative or have low proliferation rates⁷⁹, are resistant to chemotherapy and predict metastatic relapse^{80,81}. Such features have also been described for other cancer types, such as prostate and gastrointestinal cancers⁷⁷.

There is also increasing experimental evidence that supports the existence of a transient EMT–stem cell-like phenotype. In the MMTV-PyMT mouse breast cancer model, CTCs positive for the stem cell marker CD90 are responsible for metastases to the lung, but the proportion of CD90⁺ cells declines again in differentiating and growing metastases⁸². Graff *et al.*⁸³ described a dynamic DNA-methylation pattern at the E-cadherin locus of breast cancer cells *in vitro*⁸³. DNA methylation at the E-cadherin locus was increased in invading cancer cells, resulting in decreased expression of E-cadherin. Subsequent growth in three-dimensional spheres resulted in differentiation, which was characterized by reduced DNA methylation and an increase in E-cadherin expression, suggesting a high epigenetic plasticity in cancer cells. In seminal publications by Chaffer *et al.*, the importance of a re-differentiation for macrometastatic growth was first shown^{84,85}. *In vivo* selection of TSU-PR1 bladder cancer cell lines resulted in mesenchymal subclones that had a high capacity to invade, disseminate and form micrometastases, but they failed to progress to macrometastases. By contrast, subclones with an epithelial phenotype formed macrometastases after injection into the systemic circulation. Similar results were gained using an isogenic system of four breast cancer cell lines. Only the epithelial E-cadherin- and miR-200-expressing clone 4T1 formed macrometastases, whereas the mesenchymal clone 4T07, although disseminating and forming more micrometastases, did not form macrometastases. Strikingly, transfection of miR-200 into 4T07 promoted a MET and enabled macrometastatic growth⁸⁶. Recently, using the same cell culture system, Korpál *et al.*²⁸ showed that the re-expression of previously downregulated miR-200 is absolutely required for metastatic colonization. Thereby, miR-200 re-expression not only drives epithelial differentiation but also promotes macrometastatic growth by directly targeting SEC23A, which mediates the secretion of metastasis-suppressive proteins, such as insulin-like growth factor-binding protein 4 (IGFBP4) and tubulointerstitial nephritis antigen-like 1 (TINAGL1). Of note, the positive effect of miR-200 on colonization was associated with a reduced dissemination capacity, underscoring the rate-limiting role of a MET and re-differentiation for colonization in type I metastasis.

The use of *in vivo* reporter systems also supports the existence of type 1 metastases. Using TGF β -dependent reporter genes in an *in vivo* mouse model, Giampieri *et al.*⁸⁷

showed that breast cancer cells had to undergo a transient TGF β -induced EMT to form lung metastases. Constitutive TGF β signalling promoted single cell motility and dissemination, but reduced subsequent growth in the lungs⁸⁷. In a second animal model using rat prostate cancer cells, Oltean *et al.* used the fibroblast growth factor receptor 2 (FGFR2) exonIIIc, an alternative splice variant of the FGF receptor expressed only in mesenchymal cells, as a reporter gene. They showed that metastases with a differentiated, epithelial phenotype were derived from disseminated cancer cells that had initially seeded in a mesenchymal state^{88,89}. Recently, in a genetic mouse model of pancreatic cancer, it was demonstrated that prior to metastasis to the liver, cancer cells circulate and disseminate in a mesenchymal phenotype and exhibit stem cell properties⁹⁰.

Triggers of MET and re-differentiation.

Owing to the transient nature of the EMT–stem cell-like phenotype of invading and disseminating cancer cells in type I metastasis, what induces a MET in metastases is a central question. It is most likely that this is not an intrinsic tumour cell process, but one that is dependent on external and environmental factors. In this context, factors that define the metastatic niche⁹¹ should also be considered as factors that allow or induce a MET of DTCs. However, one has to keep in mind that cancer (stem) cells are genetically altered and do not respond in a similar way to external stimuli as might normal (stem) cells. Many triggers of an EMT have already been identified², however, much less is known about the inducers of a MET.

An interesting hypothesis, that epithelial differentiation is a default pathway, was proposed 15 years ago by Frisch⁹². If this were the case, EMT could only occur in the presence of positive EMT triggers, and a lack of such triggers would always result in an epithelial status. There are molecular data that partly support this hypothesis. For example, the expression of E-cadherin can stabilize a MET by sequestering β -catenin and the nuclear factor- κ B (NF- κ B) component p65, thereby inhibiting SNAIL1-induced EMT in cancer cells⁹³. These results indicate that there is a threshold level of E-cadherin for inducing a default pathway for the stabilization of an epithelial phenotype. In addition, contact of undifferentiated cancer cells with normal epithelial cells can result in their epithelial differentiation. This has been shown for prostate cancer cells⁹⁴ and for breast cancer cells in a mouse model of liver metastasis⁹⁵. There could be many

pathways involved in this process, but bone morphogenetic protein 7 (BMP7) was shown to induce a MET in renal fibroblasts⁹⁶ and in prostate and breast cancer cells^{97,98}, thereby reducing their capacity to form bone metastases. Expression of WNT inhibitory factor 1 (WIF1) in prostate cancer cells⁹⁹ and expression of FZD7 in colorectal cancer cells grown in three dimensions¹⁰⁰ can also induce epithelial differentiation.

If a MET is essential for establishing macrometastases, then an inability to undergo a MET in specific organs owing to a lack of signals might also have a role in the organ tropism seen for metastasis from different tumour types. In this context, environmental factors triggering epithelial differentiation of (dormant) DTCs or migrating cancer stem cells of different tumour entities and the co-evolution of an adequate stroma would also be factors defining their organ-specific metastatic niche. This is supported by a recent excellent study in a mouse model of breast cancer. It was demonstrated that the recruitment of bone marrow-derived myeloid progenitor cells to the premetastatic niche in the lung was essential to induce a MET of DTCs and the subsequent formation of macrometastases¹⁰¹. An active role of specific environmental factors in triggering a MET and thereby defining organ tropism of particular cancers would also argue against epithelial differentiation as a pure default pathway. Further identification of such factors might be of high importance and potential clinical relevance to prevent type I metastasis. By contrast, the formation of undifferentiated type II metastases is probably more independent of environmental stimuli and driven intrinsically, particularly by the accumulation of genetic alterations.

Genetic type II

The characteristic feature of this type of metastases is that they are undifferentiated. What is their origin and why do they not need to re-differentiate? Two principle scenarios are likely to result in undifferentiated metastases (FIG. 2). First, the primary tumour was never differentiated. It is possible that the tumour cell of origin was an early stem or precursor cell and that the tumour cells are fairly fixed in this state owing to genetic alterations (intrinsic subtype IIa). Second, the primary tumour was originally differentiated, but selected tumour subclones that have an EMT–stem cell-like phenotype are conserved in this state through rounds of genetic alterations and external selection forces (induced subtype IIb).

Intrinsic subtype. One interesting consequence of the intrinsic subtype is that a classical benign precursor lesion may not exist or may have progressed so rapidly that the primary tumour can disseminate and metastasize very early on in its development. Thus, such types of tumours would conform to the 'parallel progression' concept in which tumour and metastases progress in parallel¹⁰².

The most prominent examples for such a tumour and metastasis type are found among the triple-negative types of breast cancer. These include the histologically defined metaplastic type (also known as carcinosarcoma of the breast), as well as the basal-like and claudin-low types that have been defined based on their gene-expression signatures^{103–105}. They are highly aggressive cancers, as indicated by high rates of proliferation, chemoresistance and tumour recurrence, as well as by early and high rates of distant metastases¹⁰⁶. At the molecular level, they have an intrinsic EMT and a stem cell-like phenotype, which is characterized by the expression of SLUG, ZEB1, CD133, BMI1 and low expression of miR-200. A substantial proportion of the cells in these tumours express the cancer stem cell marker combination CD44⁺CD24^{low} (REFS 12,14,107–110). The cause of this undifferentiated EMT and stemness phenotype in the different types of triple-negative breast cancers may be different, and it is not yet known whether either specific mutations or the selection of an immature stem or progenitor phenotype are necessary, or whether both are necessary. For example, it has been suggested that the claudin-low breast cancer type is derived from more immature stem or progenitor cells than the other breast cancer types, thereby restricting its differentiation capacity¹⁰⁷. By contrast, the metaplastic type, which is characterized by biphasic histology with a carcinomatous and sarcomatous phenotype, shows a high proportion of *PIK3CA* mutations associated with EMT and stem cell-like characteristics¹⁰³. Cell lines from the basal type of breast cancers can stochastically interconvert to a luminal phenotype¹¹¹. However, the probability rate is extremely low and this transition only goes through a stem-like intermediate phenotype. Nevertheless, such data indicate that even undifferentiated phenotypes might not be completely fixed.

There are also other examples of undifferentiated types of primary carcinomas, which have lost their ability to differentiate. These include the recently defined quasi-mesenchymal subtype of pancreatic cancer¹¹². This pancreatic cancer type has an EMT–stem cell-like profile, a very poor prognosis, and

these cancer cells are resistant to standard treatment with gemcitabine and erlotinib. Endometrial carcinosarcoma, like carcinosarcoma of the breast, is a very aggressive disease and shows a true EMT phenotype, indicated by the expression of different EMT inducers and the lack of miR-200 expression¹¹³. Another example is anaplastic thyroid carcinoma, which is highly aggressive, metastatic and chemoresistant. It is characterized by a high proportion of CD133⁺ cancer stem cells, which are rapidly proliferating¹¹⁴. Moreover, in contrast to the clinically less aggressive follicular type of thyroid carcinomas, the anaplastic type has an endogenous EMT phenotype, which is characterized by high ZEB1 and low miR-200 and miR-34 expression¹¹⁵. Loss-of-function mutations in the gene encoding E-cadherin have been identified in some types of undifferentiated tumours, including the diffuse type of gastric cancer, which is characterized by early dissemination and metastasis^{116,117}. Also, this tumour type has a strong EMT phenotype, as indicated by high vimentin and SNAIL1 expression¹¹⁸.

Induced subtype. The clinically most likely and relevant cause for the switch from a differentiated to an undifferentiated tumour is treatment with rounds of chemotherapy. Tumours that recur after treatment with chemotherapy can be highly resistant, less differentiated and highly metastatic. This is supported by many clinical observations. Again, breast cancer is a prominent example. The treatment of patients with differentiated breast cancers for 3 months with conventional hormone and chemotherapy resulted in recurrent undifferentiated tumours that had an EMT and cancer stem cell-like phenotype (CD44⁺CD24^{low}), as well as an aggressive, claudin-low profile¹⁰⁸. Thus, these results support the view that selection for drug resistance in differentiated cancers results in a proposed induced genetic type IIb, which resembles the intrinsic type IIa.

Other reports have shown that cancer stem cells with an EMT phenotype remain after chemotherapy and are the sources for primary tumour recurrence and metastasis^{119,120}, indicating that chemotherapy selects for intrinsically resistant cancer stem cells. But, why do cancer stem cells of type II metastases not undergo a MET? A possible explanation could be that rounds of chemotherapy select for genetic alterations in these cells that allow both the maintenance of crucial stem cell features (particularly drug resistance and self-renewal) and sustained uncontrolled proliferation. In this case, stem

cell self-renewal would become uncoupled from stem cell quiescence, and thus re-differentiation would be unlikely and potentially unnecessary for macrometastasis. The result would be a highly proliferating, drug-resistant tumour in a permanent EMT–stem cell phenotype, which has thus already acquired all the traits for dissemination and metastatic colonization without a need to re-differentiate. Recent detection of intratumoral heterogeneity in terms of gene expression, gene mutation and genomic rearrangements indicates that many cancers can rapidly adapt to selection forces, such as chemotherapy, owing to the variety of genetic subclones present in the tumour¹²¹.

However, in this scenario, a central question remains: what are the crucial genetic alterations that keep tumours in an EMT–stem cell-like state, and in parallel allow proliferation by uncoupling stemness-associated quiescence, thus making a re-differentiation unnecessary for colonization and macrometastasis? Given the recently discovered role of p53 in controlling phenotypic plasticity by regulating the expression of miR-200 and miR-34 family members, p53 mutations may be important for maintaining an EMT stemness state. However, the high abundance of p53 mutations also in differentiated cancers, suggests that p53 mutations alone can not be sufficient, underscoring the necessity of additional mutations, such as in *CDKN2A*, to overcome quiescence or senescence. Therefore, it will be important to search for new genetic alterations that trigger type II metastasis, perhaps in cooperation with p53 mutations. In addition, if organ tropism is partly defined by the ability to induce a MET, does this mean that undifferentiated type II metastases show less tropism? A careful survey of metastases to uncommon distant sites could address this question.

Perspectives

The suggested classification of the metastatic progression into two principle types reflects the situation in ideal cases. In the clinic, however, both types might fluently overlap with a dominance of either one type or the other type. Despite this, classifying metastases into these two types allows clear links to important aspects of metastasis biology, such as the roles of environmental signals and metastatic niches versus crucial genetic alterations, organ tropism, latency and dormancy, as well as to clinical challenges, particularly overcoming treatment resistance. Defined questions and hypothetical answers can be deduced, which can be analysed in experimental and clinical settings (BOX 2).

The most obvious consequence of the proposed types of metastasis discussed in this article concerns the development of novel treatment strategies, particularly those that aim to overcome resistance to standard chemotherapy. It is now widely thought that treatment resistance involves cells that have a stem cell-like phenotype¹¹⁹. The differentiated tumour mass can often be completely eliminated by radiotherapy and/or chemotherapy, but recurrence in such patients is thought to be due to surviving cancer stem cell-like cells. For example, 12 weeks of treatment of breast cancers with neoadjuvant chemotherapy led to an enrichment of the CD44⁺CD24^{low} cancer stem cell fraction¹²², and the breast cancer cells that remained were also enriched in stem cell-like and EMT features, which is characteristic of the claudin-low subtype¹⁰⁸. The resistance of cancer stem cells compared with differentiated cancer cells might be due to the mechanisms that exist in stem cells to ensure their long-term survival and the capacity of

cancer stem cells to exist in a dormant (quiescent) state¹²³. Indeed, cancer cells that have undergone an EMT and have stem cell-like characteristics are more resistant to inducers of apoptosis and senescence, two mechanisms that are known to suppress tumour development^{2,124,125}, and many examples of an EMT-associated drug resistance in different types of cancer have been described².

Thus, the collective evidence indicates that cells with an EMT–stem cell-like phenotype in all stages of tumour progression (the primary tumour, DTCs, migrating cancer stem cells and metastases) are the major obstacle to successful cancer treatment and, therefore, are also the most important target to successfully fight metastasis¹²⁰. The increasing knowledge about the molecular links between cancer stem cell-like phenotypes and EMT and the molecular basis of phenotypic plasticity offer multiple therapeutic options (FIG. 4). However, such strategies must consider the main difference between the plasticity type I

and the genetic type II metastases — that is, the capacity to re-differentiate. This capacity allows a potential re-sensitization for conventional chemotherapy in the plasticity type I, but not in the genetic type II. Indeed, drug sensitivity can be restored by ectopic over-expression of miR-200 family members^{19,33–36}, presumably because they help to restore the differentiated state. Therefore, a promising and straightforward strategy would be to force cancer stem cells to re-differentiate. Epigenetic modifications that could be the molecular basis for plasticity type I metastasis would come into play because, in contrast to mutations, epigenetic changes can potentially be reversed by epigenetic drugs. For example, ZEB1 and SNAIL repress endogenous expression of *mir-200* or *mir-34* genes by recruiting epigenetic co-repressors, such as histone deacetylases (HDACs) and the epigenetic regulator lysine-specific demethylase 1 (LSD1)^{22,126,127}. Epigenetic repression of miR-200 by ZEB1 is reversible and can be

Box 2 | Open questions

There are several important questions that arise from the classification of type I and type II metastasis that need to be answered. Some hypothetical answers are suggested below.

Plasticity

When do the changes that enable phenotypic plasticity occur?

Because phenotypic plasticity is already evident in differentiated primary tumours, the selection forces that enable this feature in the primary tumour must be present before dissemination and metastasis occur. Consequently, phenotypic plasticity not only favours metastasis but is also already supportive of invasion and the growth of primary tumours.

What are the main differences between normal cancer stem cell traits and cancer-associated phenotypic plasticity?

The underlying genetic alterations in cancer cells might change the reaction of the cell to external signals, such as those from stem cell niches and infiltrating cells, and limit its differentiation capacity. Consequently, phenotypic plasticity in cancer may not necessarily result in a strictly hierarchical organization, but cancer cells might be able to interconvert between their phenotypic states: that is, non-cancer stem cells could produce cancer stem cells.

Is epithelial–mesenchymal transistion (EMT) the right term?

EMT is a well-established term that is restricted to the mesenchymal transition of epithelial cells. However, EMT activators, such as SNAIL factors and ZEB factors, also induce stem cell-like properties escape from fail-safe mechanisms and resistance to apoptosis; therefore, the induction of EMT may only be one aspect of their different functions. Dedifferentiation (the stepwise transition towards a stemness phenotype) may be a broader definition of their function. This also takes into account the fact that EMT activators are expressed in and induce dedifferentiation of non-epithelial cells, such as glioma and melanoma cells.

Genetic background

Why do some differentiated tumours with identical histological grading undergo an EMT at the invasive front (type 1a, which is associated with high metastatic capacity) and others do not (type 1b)?

These differences might be explained by the genetic background. Genetic variances could lead to different expression levels of EMT-inducing stimuli,

perhaps owing to different inflammation and infiltration patterns at the invasive front (the interface between the tumour and the host tissue), and may be identified by genetic linkage analyses.

Organ tropism and dormancy

What triggers a mesenchymal–epithelial transition (MET) in type I metastasis?

Signals inducing MET in metastases are mostly unknown and may be crucial components of a successful metastatic niche. Because re-differentiation is important for colonization and macrometastasis, such signals, if they differ between various cancer types, are likely to also be involved in defining the organ tropism of carcinomas.

Why do carcinomas rarely metastasize into mesenchymal tissues (such as skeletal muscle)?

Type I carcinomas only metastasize to distant sites allowing re-differentiation. Physiological stem cell niches in mesenchymal tissues do not support epithelial differentiation; therefore, the development of an adequate metastatic niche allowing the induction of a MET in disseminated cancer cells is unlikely. If true, (rare) metastasis of carcinomas into mesenchymal tissues should mostly result from type II metastases, as they do not need to re-differentiate.

Which genetic alterations uncouple the EMT–stem cell-like phenotype from the associated growth arrest and quiescence of stem cells, allowing the permanent proliferation of undifferentiated type II metastases?

In addition to the proposed shift towards an EMT state, p53 mutations may participate in the loss of EMT–stem cell-like quiescence, although this is unlikely to be the only mechanism. Additional genetic alterations that cooperate with known mutations may exist that uncouple proliferation from the postulated fixed EMT–stemness phenotype of type II metastases. Such alterations should be identified in the current programmes of whole-cancer genome sequencing.

Is dormancy only relevant (and possible) in differentiated type I cancers, and are type II tumours not dormant?

Owing to the intrinsic and genetic uncoupling of proliferation and quiescence, type II tumours and their derived DTCs can probably not acquire a dormancy state, which would also explain an earlier and higher risk for recurrence and metastasis.

achieved by applying epigenetically active drugs, such as the DNA-demethylating agent 5-aza-2'-deoxycytidine or HDAC inhibitors^{39,128}. Because re-expression of miR-200 restores chemosensitivity and radiosensitivity in cancer stem cells, pretreatment of tumours with epigenetic drugs before standard chemotherapy might prevent tumour recurrence, metastasis formation and might even target existing metastases and DTCs.

An alternative strategy would be to prevent cancer cells from undergoing a MET, which, hypothetically, would keep DTCs and micrometastasis in a quiescent EMT–stem cell-like state or in a dormant state. The identification of metastatic niche signals that are necessary to induce a MET would offer relevant targets to exert such a strategy. But a major disadvantage would be the need for chronic and potentially lifelong treatment.

However, both strategies would only be an option for the plasticity type I metastasis. Genetic type II metastases can probably not be induced to differentiate and to restore chemosensitivity. In addition, if one postulates a high proliferation rate, genetically uncoupled from the quiescent stem cell state, a strategy to induce dormancy by maintaining this state would also fail. Thus, the only way to treat type II metastases is to directly target

the EMT–stem cell state. The development of drugs that selectively target cancer stem cells would benefit both types of metastasis, but this is a challenging task. *In vitro* and preclinical treatment studies are currently being carried out to find combinations of drugs that selectively target signalling pathways that are active in stem and progenitor cells, such as the WNT¹²⁹, Hedgehog (HH)¹³⁰, AKT–mTOR^{131,132} and Notch pathways^{133,134}. Metformin, a first-line drug used for treating type II diabetes, was reported to selectively kill a chemoresistant subpopulation of cancer stem cells in an *in vivo* model of breast cancer, although the molecular mechanism of its action and selectivity is unknown^{135,136}. Another example is dasatinib, an orally active inhibitor of both SRC and ABL kinases that can preferentially inhibit the growth of breast cancers with an EMT–stem cell-like phenotype, particularly triple-negative cancers of the basal subtype¹³⁷. Unbiased screening programmes for drugs that selectively kill cancer cells with an EMT–stem cell-like phenotype should also lead to promising new therapies. Such an approach has identified the drug salinomycin, which selectively acts on cancer stem cells¹³⁸.

However, a major problem for future therapeutic strategies may arise from principle

differences in the biology of normal stem cells and cancer stem cells. The molecular basis of these differences is not well understood and is likely to be due to the underlying genetic alterations that disrupt the normal hierarchy of stem, progenitor and differentiated cells and make tumour cells much more flexible in interconverting between these states. An important consequence is that cancer stem cells might arise again from non-cancer stem cells, by activating an EMT programme^{111,139,140}. A dynamic shift between a drug-resistant and a drug-sensitive state of cancer cells, mediated by epigenetic changes, has been reported¹⁴¹. Thus, do we need to target all types of cancer cell subpopulations (cancer stem cells, CTCs, DTCs and differentiated cancer cells) at the same time? Such concerns indicate that any single agent therapy is likely to fail and further underscore the importance of developing combination therapies that target different steps in the metastatic cascade.

Conclusions and future directions

Accumulating data indicate that rounds of transient transitions between EMT–stem cell-like phenotypes and MET–re-differentiation phenotypes in tumour cells are the basis for dissemination and metastasis of differentiated

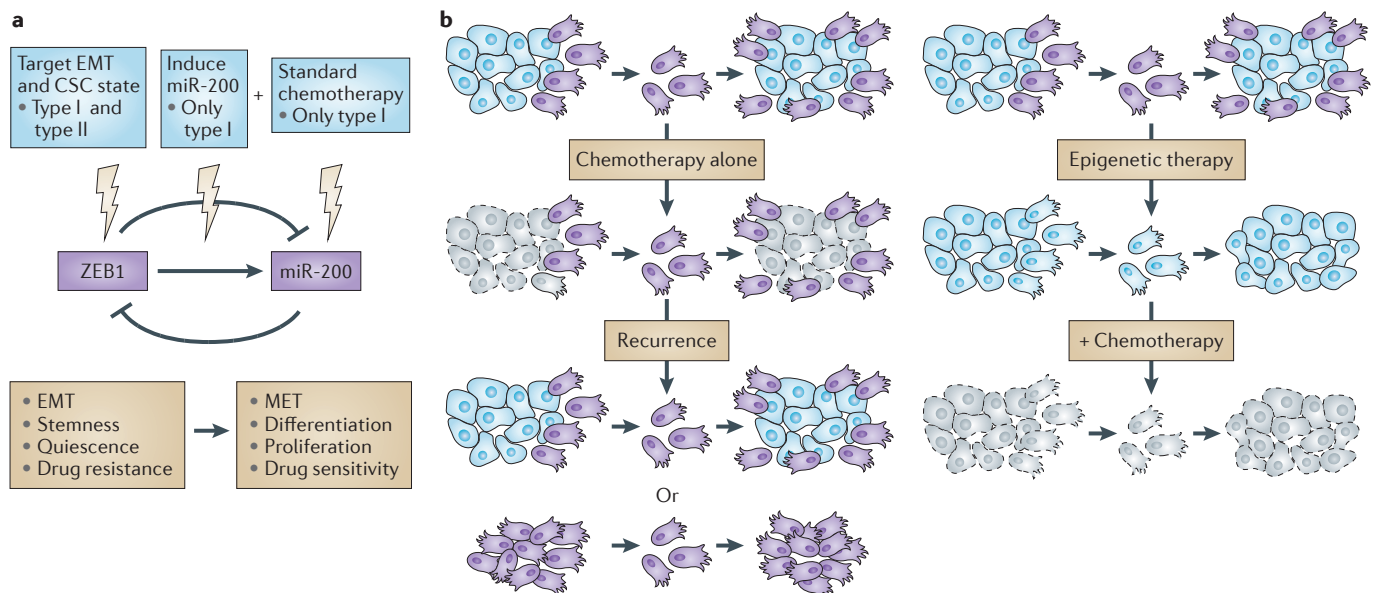


Figure 4 | Therapeutic strategies against metastasis. **a** | Different treatment strategies, targeting either the stemness phenotype or interrupting phenotypic plasticity, exemplified by the ZEB–miR-200 feedback loop, are shown. Tumour cells of primary tumours, metastases and disseminating tumour cells (DTCs) in an epithelial–mesenchymal transition (EMT)–stemness state can be directly targeted by specific drugs, such as metformin. This might be applied to both types of metastases, eventually in combination with standard chemotherapy. In addition, plasticity itself can be a target. This can be achieved by applying epigenetically active

drugs, such as new generations of histone deacetylase (HDAC) inhibitors, which induce differentiation, possibly by re-activating the silenced expression of miR-200. The associated re-acquisition of drug sensitivity allows a parallel combination therapy with standard chemotherapeutics or irradiation. This strategy is only applicable to type I tumours, as type II tumours have a low re-differentiation capacity. **b** | Schematic illustration of the effects of chemotherapy alone or in combination with epigenetic therapy on type I tumours (dashed lines indicate successfully targeted phenotypes). CSC, cancer stem cell; MET, mesenchymal–epithelial.

carcinomas. In these tumours it is not the fixation in one phenotype that favours metastatic progression, but it is the aberrant ability of the tumour cell to switch from one state to the other that allows permanent adaptations to the demanding conditions of a changing environment. Thus, phenotypic plasticity is the crucial molecular trait of metastasis type I produced by differentiated cancers. This also underscores a dominant role for environmental and contextual influences on this process. Phenotypic plasticity is not evident in undifferentiated cancers, in which genetic alterations are proposed to be the major driving force for genetic type II metastasis.

However, there is still no definitive proof for the existence of transient EMT and MET processes, and this might never be gained for human metastatic cancer. More pre-clinical evidence is required, and this could be achieved by lineage tracing of an EMT phenotype in mouse models of metastatic cancer. In addition, studies of CTCs and DTCs, directly isolated from human cancers, should not only include genomic and gene expression profiling, but also *in vivo* functional analyses to determine their phenotypic plasticity and metastatic capacity. Moreover, environmental signals in the metastatic niche that can induce differentiation of seeded DTCs should be identified, which would also offer new therapeutic targets. Furthermore, it will be of prime interest to identify genetic alterations that maintain type II metastasis-prone cancers in an undifferentiated EMT state and also those that allow a simultaneous high proliferation rate. Current programmes of sequencing whole-cancer genomes should also consider identifying such genetic alterations. A topic not touched in this Opinion article is the metastasis of non-epithelial (mesenchymal) cancers. Do EMT-like processes also have a role in the metastasis of sarcomas, and, if so, how would this fit with the physiological role of EMT inducers? Finally, it would be clinically relevant to develop diagnostic and predictive biomarkers that would allow the prediction of the most likely type of metastasis on the basis of information available in the primary tumour. A prediction of the likely type of metastasis would also be important for designing clinical trials to assess novel and specific treatment strategies.

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FURTHER INFORMATION

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OPINION

Towards the use of cannabinoids as antitumour agents

Guillermo Velasco, Cristina Sánchez and Manuel Guzmán

Abstract | Various reports have shown that cannabinoids (the active components of marijuana and their derivatives) can reduce tumour growth and progression in animal models of cancer, in addition to their well-known palliative effects on some cancer-associated symptoms. This Opinion article discusses our current understanding of cannabinoids as antitumour agents, focusing on recent insights into the molecular mechanisms of action, including emerging resistance mechanisms and opportunities for combination therapy approaches. Such knowledge is required for the optimization of preclinical cannabinoid-based therapies and for the preliminary clinical testing that is currently underway.

Few plant species have been the subject of so much scientific, clinical and social debate as *Cannabis sativa* L. (marijuana). Preparations from this plant have been used for many centuries both medicinally and recreationally. However, the chemical structures of their unique active components — the cannabinoids — were not elucidated until the 1960s. Three decades later, the first solid clues on cannabinoid molecular action were established, which led to an impressive expansion of basic cannabinoid research and to a renaissance in the study of the therapeutic effects of cannabinoids in various fields, including oncology.

Today, it is widely accepted that, of the ~70 cannabinoids produced by *C. sativa*, Δ^9 -tetrahydrocannabinol (THC) is the most relevant owing to its high potency and abundance in plant preparations^{1,2}. THC exerts a wide variety of biological effects by mimicking endogenous substances — the so-called endocannabinoids (the two most studied being anandamide³ and 2-arachidonoylglycerol (2-AG)^{4,5}) — that engage specific cell-surface cannabinoid receptors⁶ (FIG. 1).

So far, two major cannabinoid-specific receptors — CB1 and CB2 — have been cloned and characterized from mammalian tissues^{7,8}. In addition, other receptors, including the transient receptor potential cation channel subfamily V member 1 (TRPV1) and certain orphan G protein-coupled receptors, GPR55, GPR119 and GPR18, have been proposed to act as endocannabinoid receptors⁶. Most of the effects that are produced by cannabinoids in the nervous system and in non-neural tissues rely on CB1 receptor activation. Expression of this receptor is abundant in the central

nervous system, particularly in discrete areas that are involved in the control of motor behaviour (such as the basal ganglia and cerebellum), memory and learning (the cortex and hippocampus), emotions (the amygdala), sensory perception (the thalamus), and autonomic and endocrine functions (the hypothalamus, pons and medulla). In addition, CB1 receptors are expressed in peripheral nerve terminals and in many extra-neural sites. By contrast, the CB2 receptor was initially described as present in the immune system⁶, but more recently it has also been shown to be expressed in additional cell types^{9,10}. Notably, expression of CB1 and CB2 receptors has been found in many types of cancer cells, although this does not necessarily correlate with the expression of these receptors in the tissue type of origin^{9,11,12}.

The endocannabinoids, together with their receptors and the proteins that are responsible for their synthesis, transport and degradation, constitute the endocannabinoid system. Aside from its pivotal neuromodulatory activity¹³ (FIG. 1), the endocannabinoid system exerts other regulatory functions in the body, such as the control of cardiovascular tone, energy metabolism, immunity and reproduction^{14,15}. This miscellaneous activity makes the pharmacological manipulation of the endocannabinoid system a promising strategy for the management of many different diseases. Specifically, cannabinoids are well-known to exert palliative effects in cancer patients^{14,15}, and their best-established use is in the inhibition of chemotherapy-induced nausea and vomiting^{15,16}. Today, capsules of THC, named dronabinol (Marinol; Solvay Pharmaceuticals), and its synthetic analogue